AI Assignment: Classical Reasoning Agent

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1.Representation

All sentences are represented using first-order Horn logic. A literal is (negated, predicate,arguments). Facts are ground literals, and rules are body->head, where body is a conjunction of literals. Pronouns and words like someone, something, people, and things turn into variables (?X,?Y), and the names turn into constants. The parser uses regular expressions and takes care of:simple facts (Bob is young), negated facts (Bob is not kind), relations (The cat chases the dog),if-then rules with conjunctions, generic rules like "Big red things are nice". The rule heads can benegative literals, and sentences not understandable by the parser are simply ignored and counted.

2. Reasoning algorithms

The inference is performed using forward-chaining. The body of each rule is unified against the fact base,with backtracking in the case of variable bindings. Negation-as-failure (closed-world assumption) is handled via stratification, meaning that before any rule using negation is applied, its negated predicate must be completely derived. The fact base uses the index (sign, predicate). Prior to inference, the backward-relevance analysis is
performed, which prunes all rules which cannot help with deriving the query predicate. Each fact contains the rule and the premises which have lead to derivation of this fact; therefore, the proof traces are available out of the box.A query is True if the literal can be derived; a negated query is also True if its positive atom cannot be derived (CWA).

3. Experimentation with public dataset

demo | CWA=True | 10 queries | accuracy 100.00% | 0.001 s
demo | CWA=False | 10 queries | accuracy 90.00% | 0.000 s
I experimented with the closed world and open world interpretation of 'not' and chose the version which has higher average accuracy (CWA = True). The final public accuracy is 100.00%, while the total time is 0.001 s.
Limitations: since my approach is rule based, there is no support for free form FOLIO sentences involving quantifiers, disjunctions and other unusual constructions, apart from CounterBench causal phrasing which can be transformed into if then statements. No experiment was done with any private test data.

4. Reasoning trace examples

Example A (from the problem sheet): Bob is young, Bob is kind; young implies smart; smart and kind implies successful. Query: Is Bob successful?
successful(bob) <- someone who is both smart and kind is successful
smart(bob) <- someone who is young is smart
young(bob) <- [Given]

kind(bob) <- [Given]
Example B (negation as failure): Fiona is young; Erin is young and kind; young but not kind implies rude.
Query: Is Fiona rude?
rude(fiona) <- someone who is both young and not kind is rude
young(fiona) <- [Given]
not kind(fiona) <- [cannot be derived]